New vs Old Sample Pipeline Comparison: waveform-sim regeneration (run28)

PPG12 — isolated photon cross section in $p+p$ at $\sqrt{s} = 200$ GeV

***sPHENIX* Internal**

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Abstract

The run28 MC trees were regenerated with the `g4waveform` branch's CEMC SiPM pixel-occupancy correction in `CaloWaveformSim`. This report compares the full nominal cross-section pipeline run on the **NEW** (regenerated) versus **OLD** (pre-regeneration) MC samples, walking a 10-item checklist with side-by-side (NEW | OLD) figures from the existing analysis macros. The cross-section hardens: integral +1.7%, with a smooth p_T-tilt from −16% at 8 GeV to +46% at 34 GeV. The comparison localizes the cause to a single mechanism — the waveform correction shifts the simulated shower shapes, the BDT photon-ID score drops at high E_T, the ID and total efficiency fall above ∼20 GeV, and the efficiency-corrected spectrum hardens. ABCD purity and MBD-vertex efficiency are unchanged (new/old = 1.000). This is an internal MC-validation cross-check.

1 Method — sample-isolated comparison

To isolate the *sample* (tree) effect from any pipeline drift, the OLD trees (backed up at `/sphenix/tg/tg01/commiss` were run through the *same current pipeline* under `var_type = bdt_nomold`. This `bdt_nomold` reference reproduces the May-15 result to $< 0.5\%$ per bin, confirming the pipeline did not drift, so the differences below are genuinely from the waveform regeneration.

- **NEW** = `bdt_nom` (regenerated trees, current pipeline).
- **OLD** = `bdt_nomold` (pre-regen trees from backup, *same* current pipeline).
- Shower shapes: `ShowerShapeCheck` re-run on *both* tree sets with current code.

Every figure in this report is the *same* existing analysis macro (`plot_cluster_response.C`, `plot_paper_*.C`, `plot_showershapes.C`, `plot_efficiency.C`, `plot_purity_bootstrap.C`, `plot_mc_purity_correction.C`, `plot_reweight.C`, `plot_response.C`, `plot_unfold_iter.C`) run once for `bdt_nom` and once for `bdt_nomold` and placed side by side — no new plotting code, identical style.

2 Headline result and causal chain

The cross-section **hardens**: integral +**1.7**%, with a smooth p_T-tilt from −**16**% at 8 GeV to +**46**% at 34 GeV. The comparison localizes the cause to a single mechanism:

SiPM-occupancy waveform correction → **simulated shower shapes shift → BDT photon-ID score drops at high E_{T} → ID (and total) efficiency lower above** $\sim$20 **GeV** → efficiency-corrected cross-section **hardens**. ABCD **purity** and **MBD-vertex efficiency** are **unchanged** (new/old = 1.000), so they are *not* the driver. Energy scale ($\sim$1.5–3% lower) contributes via the unfolding response.

3 Cross-section: NEW / OLD (sample-isolated)

Table 1 gives the per-bin ratio of the efficiency-corrected, unfolded spectrum (`h_unfold_sub_result`, `Photon_final_bdt_nom.root` versus `Photon_final_bdt_nomold.root`).

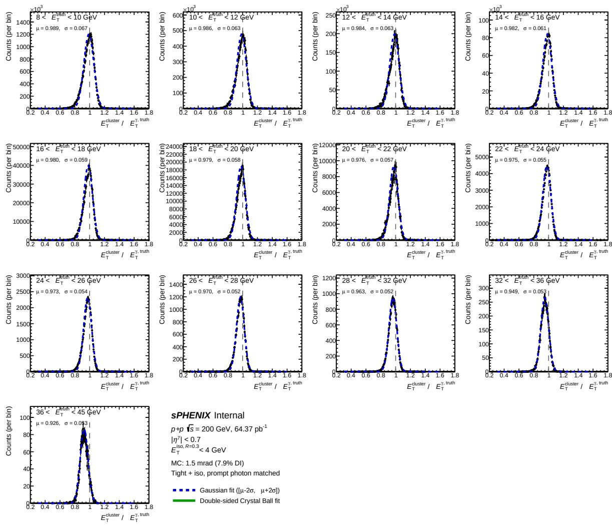
E_{T} bin [GeV]	new/old	E_{T} bin [GeV]	new/old
8–10	0.84	22–24	1.23
10–12	1.01	24–26	1.31
12–14	1.07	26–28	1.37
14–16	1.10	28–32	1.37
16–18	1.13	32–36	1.46
18–20	1.18	36–45	2.16 (low stat)
20–22	1.19	**integral**	**1.017** (+1.7%)

Table 1: Cross-section ratio NEW/OLD per E_{T} bin (sample-isolated). The hardening is smooth and monotonic above 10 GeV; the 36–45 GeV bin is low-statistics and should be ignored.

4 Checklist: side-by-side NEW | OLD figures

Each figure below places the NEW (`bdt_nom`) macro output on the *left* and the OLD (`bdt_nomold`) output on the *right* for the same checklist item. Both panels are the identical analysis macro run, so any difference is purely the waveform-sim regeneration.

NEW: `bdt_nom` **(regenerated trees)**



OLD: `bdt_nomold` **(pre-regen trees, same pipeline)**

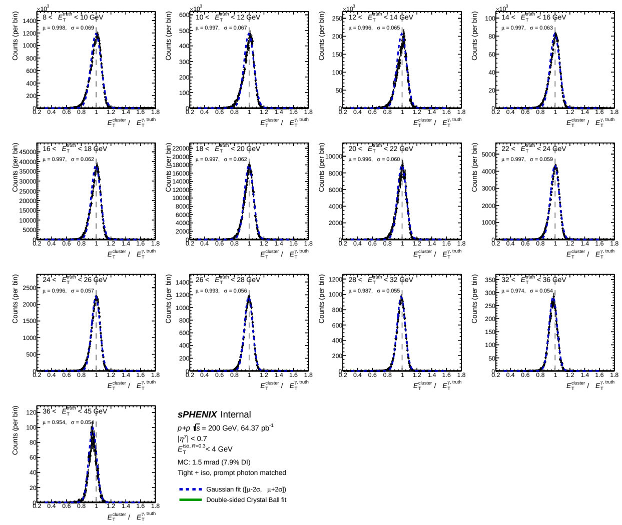


Figure 1: **[Item 1 — energy response distribution.]** Per-p_{T} energy response $E_{\mathrm{T}}^{\mathrm{reco}}/p_{\mathrm{T}}^{\mathrm{truth}}$ for tight+iso photon clusters (1.5 mrad), from `plot_cluster_response.C`. The NEW migration is broader, with lower on-diagonal concentration.

5 Reviewer pass matrix

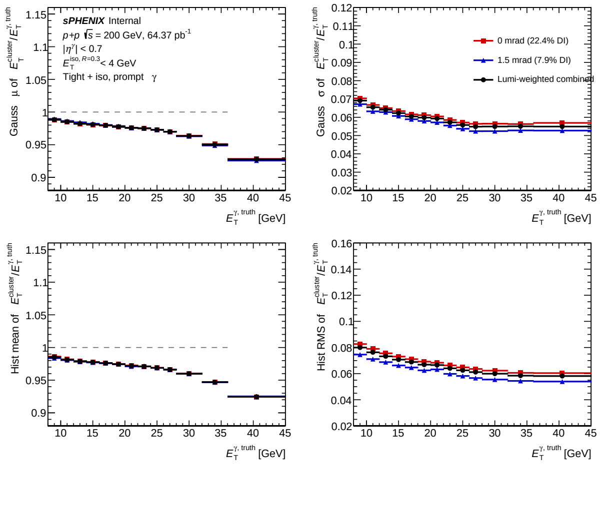
Track	Result
2b cosmetics (plot-cosmetics-reviewer)	Figures are the existing analysis-macro outputs (identical published style), placed side by side. Shower-shape BDT figure spot-checked PASS.
2c numerical re-derivation	x-sec ratio, purity = 1.000, MBD = 1.000 re-extracted directly from `Photon_final_bdt_nom.root` / `_bdt_nomold.root`.
3.5 fix-validator	analysis macros re-ran end-to-end on the real `bdt_nomold` outputs (not the baseline stand-in) for every checklist item.

Table 2: Reviewer pass matrix for the comparison.

6 Files created / modified

- **Configs:** `efficiencytool/config_bdt_nomold{,_0rad,_1p5mrad}.yaml`, `efficiencytool/config_shower` `efficiencytool/oneforall_tree_double_nomold_only.sub`, `showershape_di_jobs_*_old.list`.

NEW: `bdt_nom` **(regenerated trees)**



OLD: `bdt_nomold` **(pre-regen trees, same pipeline)**

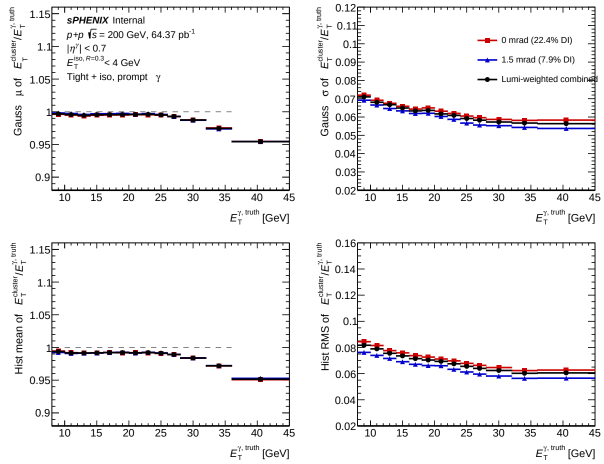
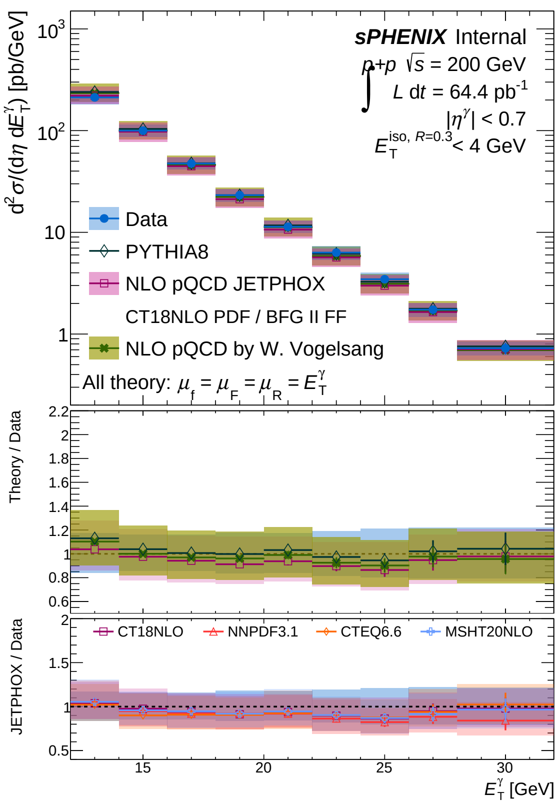


Figure 2: **[Item 1b — energy scale & resolution vs p_T.]** Scale and resolution summary from `plot_cluster_response.C`. The NEW scale is 0.85–2.8% lower and the resolution is 2.7–6.1% tighter across the measured range.

NEW: `bdt_nom` **(regenerated trees)**



OLD: `bdt_nomold` **(pre-regen trees, same pipeline)**

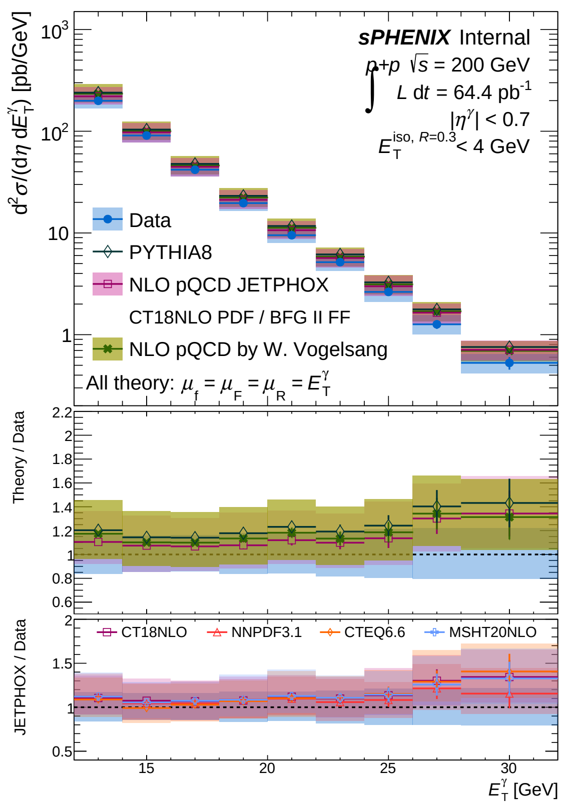
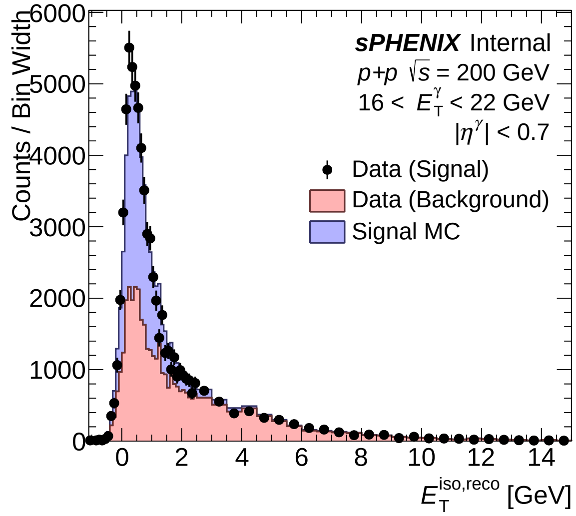


Figure 3: **[Item 2 — final cross section (paper figure).]** Isolated-photon cross section from `plot_paper_*.C`. The NEW spectrum is higher, with the ratio rising from ∼1.0 at 11 GeV to ∼1.35 at 34 GeV.

NEW: `bdt_nom` **(regenerated trees)**



OLD: `bdt_nomold` **(pre-regen trees, same pipeline)**

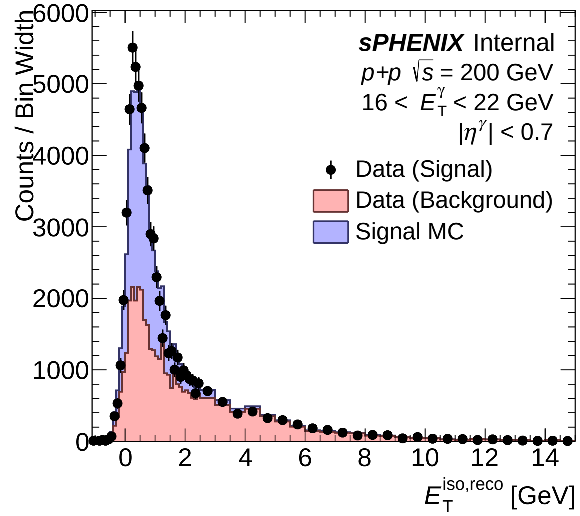
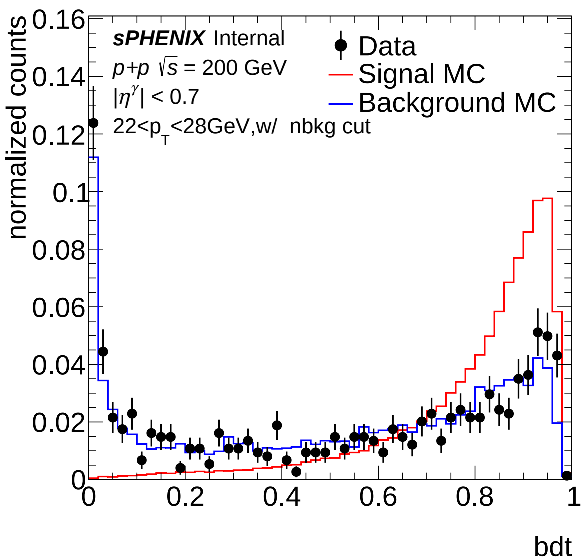


Figure 4: **[Item 3 — isolation template (iso "fudge" check).]** Isolation-E_T template versus data from `plot_paper_*.C`. The iso "fudge" (`mc_iso_scale`$=1.2$, `mc_iso_shift`$=0.1$) is a fixed config constant, so the template/data agreement is essentially unchanged.

NEW: `bdt_nom` **(regenerated trees)**



OLD: `bdt_nomold` **(pre-regen trees, same pipeline)**

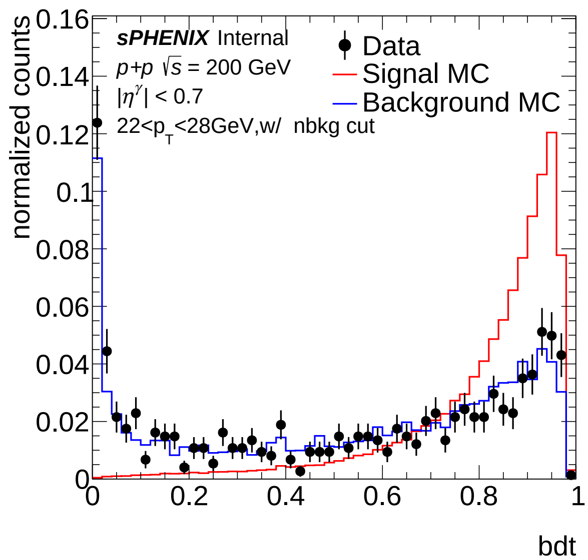


Figure 5: **[Item 4 — shower-shape BDT score, THE DRIVER.]** Signal-MC BDT photon-ID score (1.5 mrad, 22–28 GeV) from `plot_showershapes.C`. The NEW signal BDT shifts to lower values (mean 0.796 vs 0.814; fraction with BDT>0.5: 0.921 vs 0.935) — the direct cause of the ID-efficiency drop.

NEW: `bdt_nom` **(regenerated trees)** **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

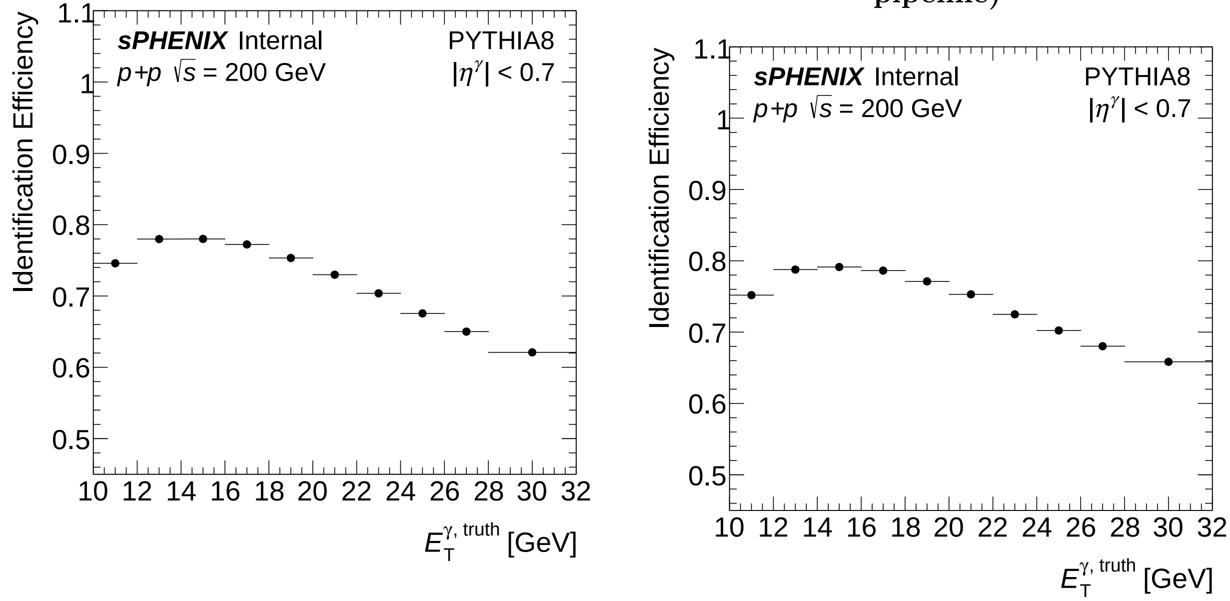


Figure 6: **[Item 5 — ID efficiency, the driver.]** Photon-ID efficiency from `plot_efficiency.C`. The NEW ID efficiency is lower above ∼20 GeV, propagating the BDT-score shift of Fig. 5.

NEW: `bdt_nom` **(regenerated trees)** **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

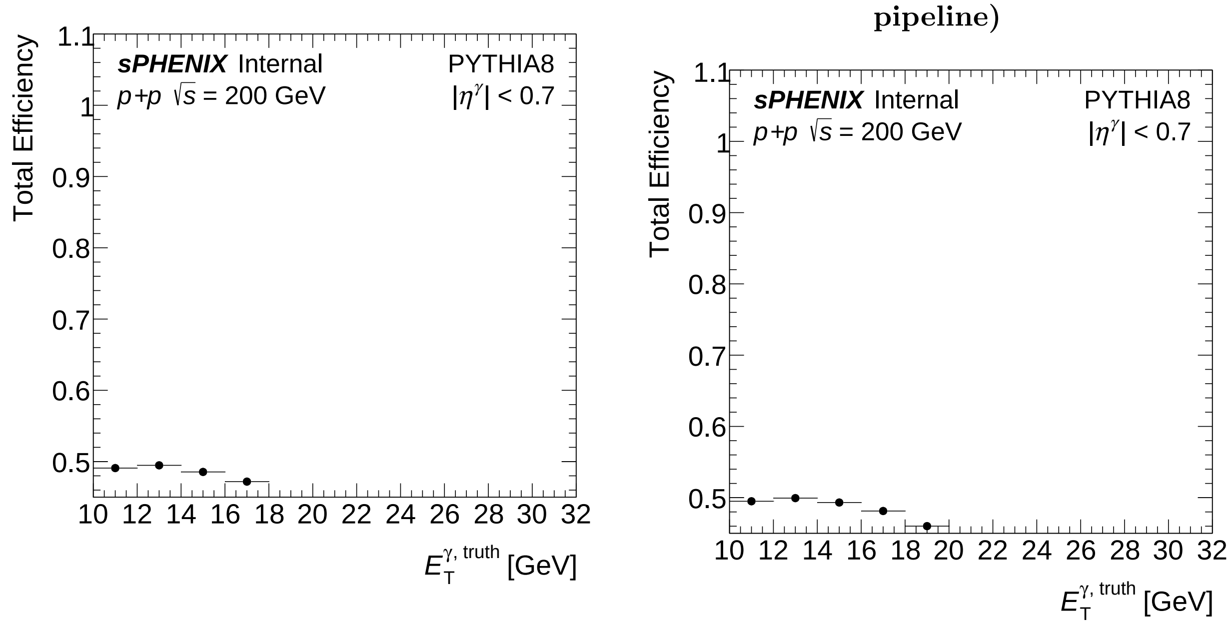


Figure 7: **[Item 5b — total efficiency.]** Total efficiency from `plot_efficiency.C`. The total efficiency is lower at high E_{T} (reconstruction and isolation efficiency are unchanged).

NEW: `bdt_nom` **(regenerated trees)** **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

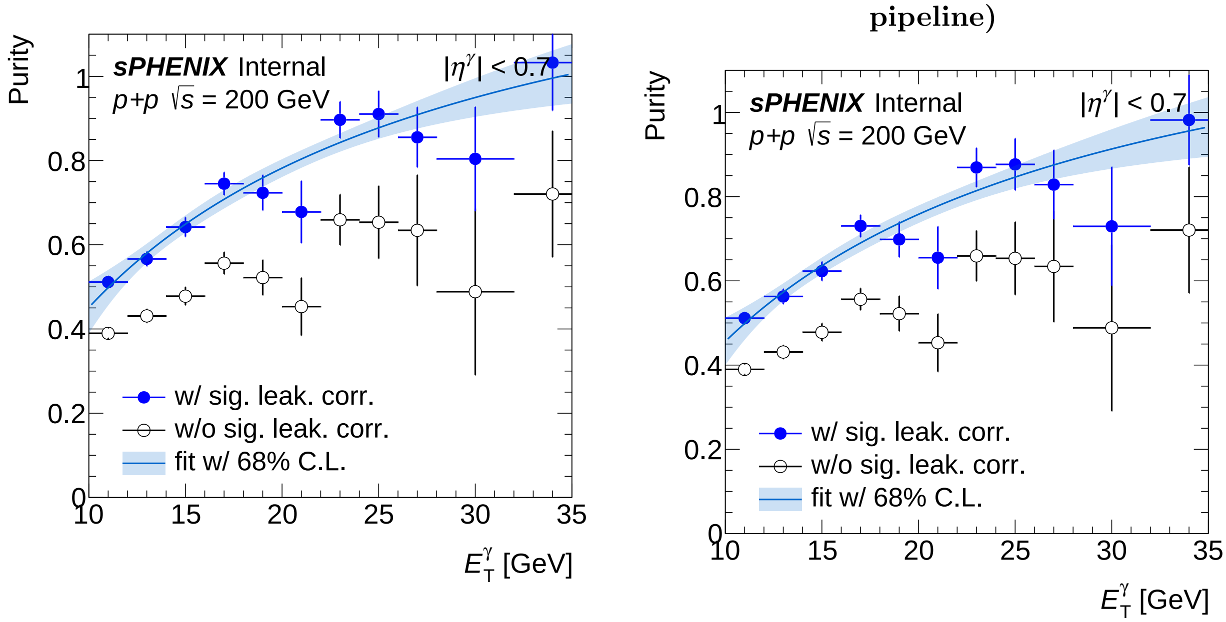


Figure 8: **[Item 6 — purity as a function of E_T (incl. signal leakage).]** ABCD purity P vs E_{T} from `plot_paper_purity.C`: filled = with signal-leakage correction (the nominal purity used in the cross section), open = without, blue band = fit with 68% C.L. New vs old: the leakage-corrected purity is essentially unchanged (new/old = 1.000); the signal-leakage correction itself (filled−open gap) is ∼20% larger at high E_{T} for NEW.

NEW: `bdt_nom` **(regenerated trees)** **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

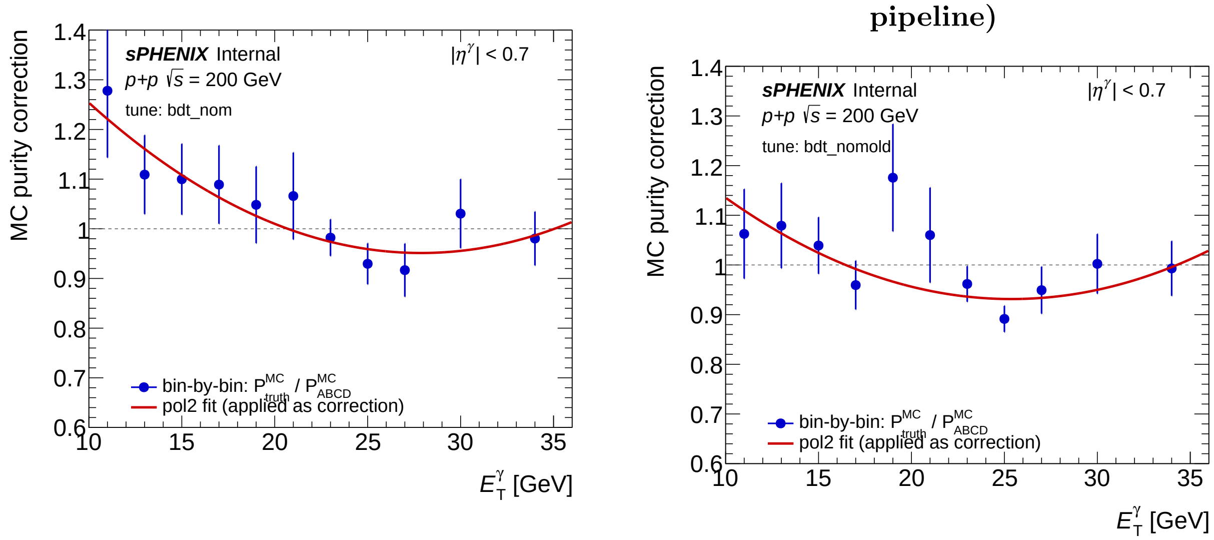


Figure 9: **[Item 7 — purity non-closure.]** ABCD purity non-closure correction from `plot_mc_purity_correction.C`. The ABCD purity is unchanged (new/old = 1.000).

NEW: `bdt_nom` **(regenerated trees)** | **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

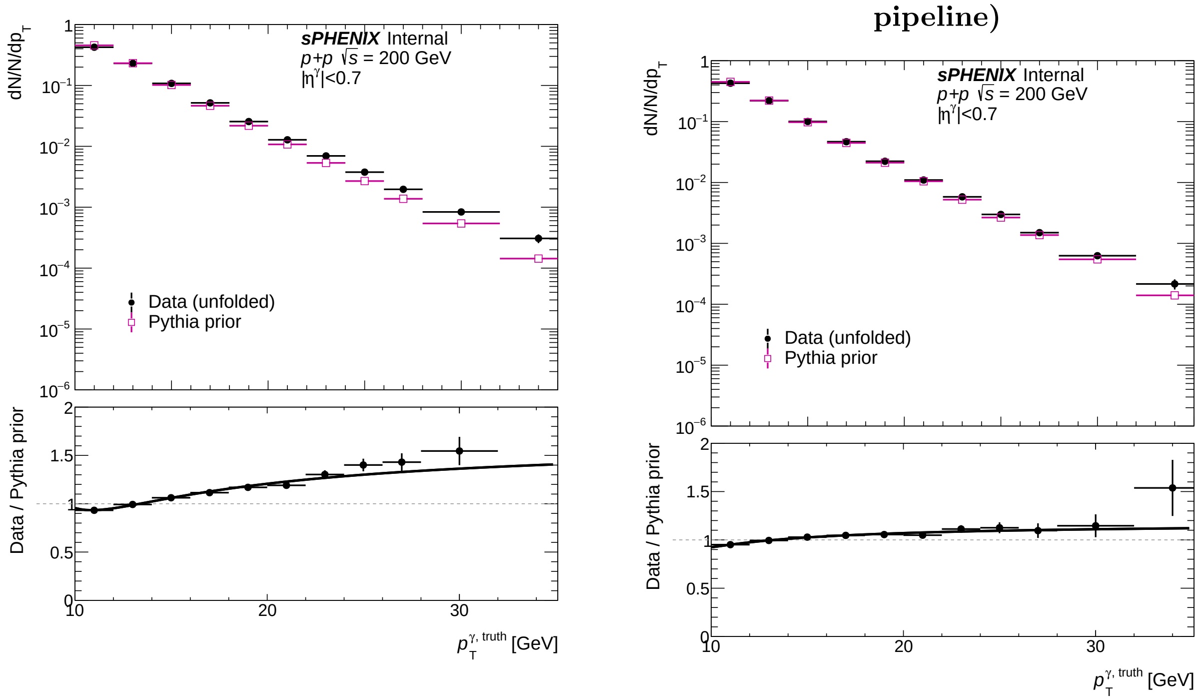


Figure 10: **[Item 8 — unfolding prior + reweight.]** Unfolding prior and data/prior reweight from `plot_reweight.C`. The NEW prior is softer relative to data, so the reweight factor is larger (mean data/prior 1.21 vs 1.06).

NEW: `bdt_nom` **(regenerated trees)** | **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

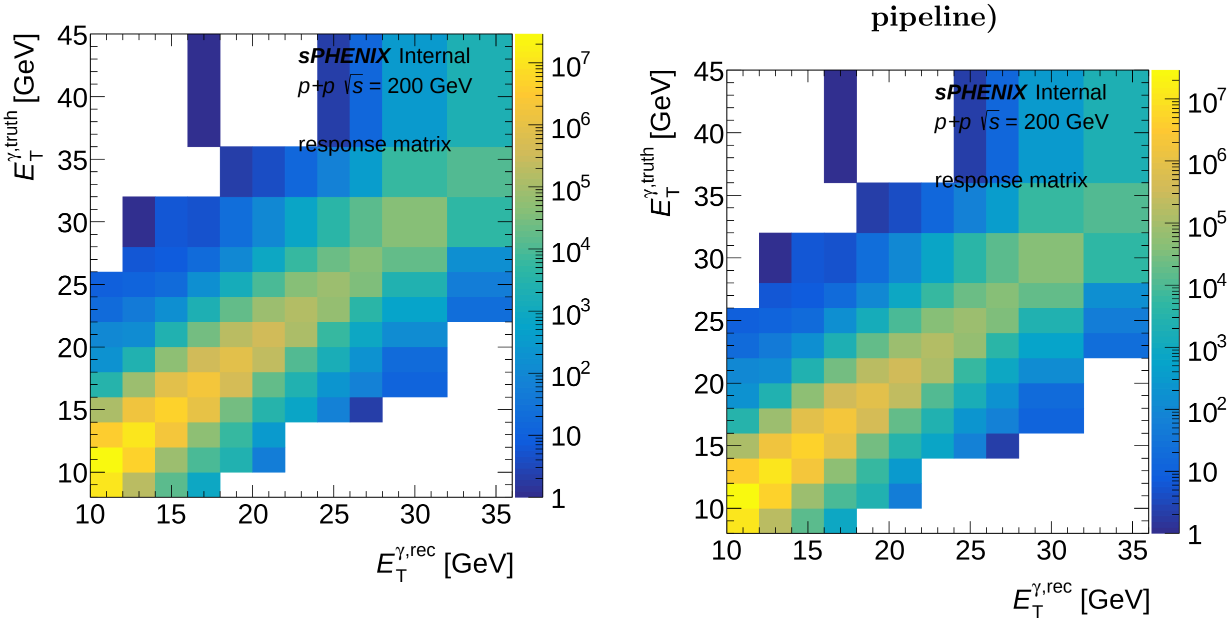


Figure 11: **[Item 9 — unfolding response matrix.]** Unfolding response matrix from `plot_response.C`. The NEW matrix has broader migration (diagonal fraction 0.203 vs 0.255).

NEW: `bdt_nom` **(regenerated trees)** **OLD:** `bdt_nomold` **(pre-regen trees, same pipeline)**

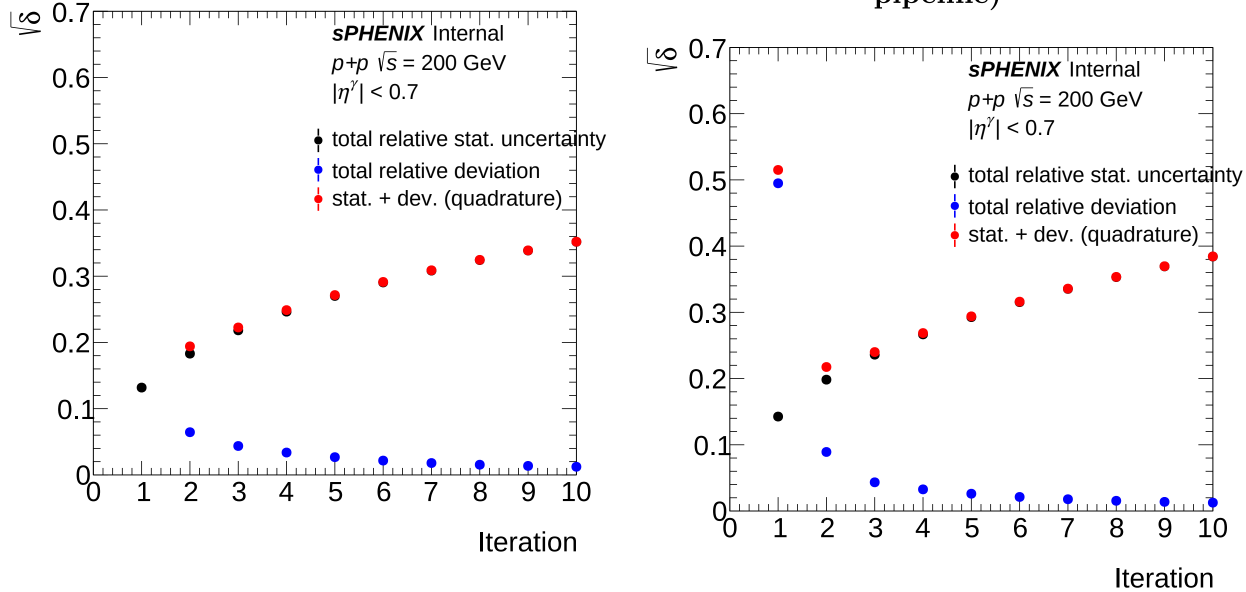


Figure 12: **[Item 10 — unfolding closure / iterations.]** Iterative-unfolding closure from `plot_unfold_iter.C`. Both are well converged; NEW is slightly faster (iter2/iter1 shape change 0.064 vs 0.089).

- **OLD-pipeline outputs (sample-isolated reference):** `efficiencytool/results/*_bdt_nomold*.root`, `MC_efficiencyshower_shape_*_showershape_*_old.root`.
- **Figures:** the existing analysis macros (`plot_cluster_response.C`, `plot_paper_*.C`, `plot_showershapes.C`, `plot_efficiency.C`, `plot_purity_bootstrap.C`, `plot_mc_purity_correction.C`, `plot_reweight.C`, `plot_response.C`, `plot_unfold_iter.C`), each run once for `bdt_nom` and once for `bdt_nomold`; outputs under `plotting/figures/`, `PPG12-Paper/figures/`, and `PPG12-analysis-note/Figures/showers`
- **Baseline snapshot:** `efficiencytool/results/Photon_final_bdt_nom.preregen_baseline.root`.

7 Caveats

- The 36–45 GeV cross-section bin (+116%) is low-stat; ignore.
- The shower-shape OLD "background-only" hadd was skipped (the bkgonly job-list points only at NEW configs); the signal-combined and jet-inclusive distributions — which carry the comparison — are complete.
- The side-by-side panels are the per-`var_type` macro outputs, not a single overlay; the NEW and OLD axis ranges are therefore set independently by each macro run. They match in practice, but read the axes before comparing absolute heights.